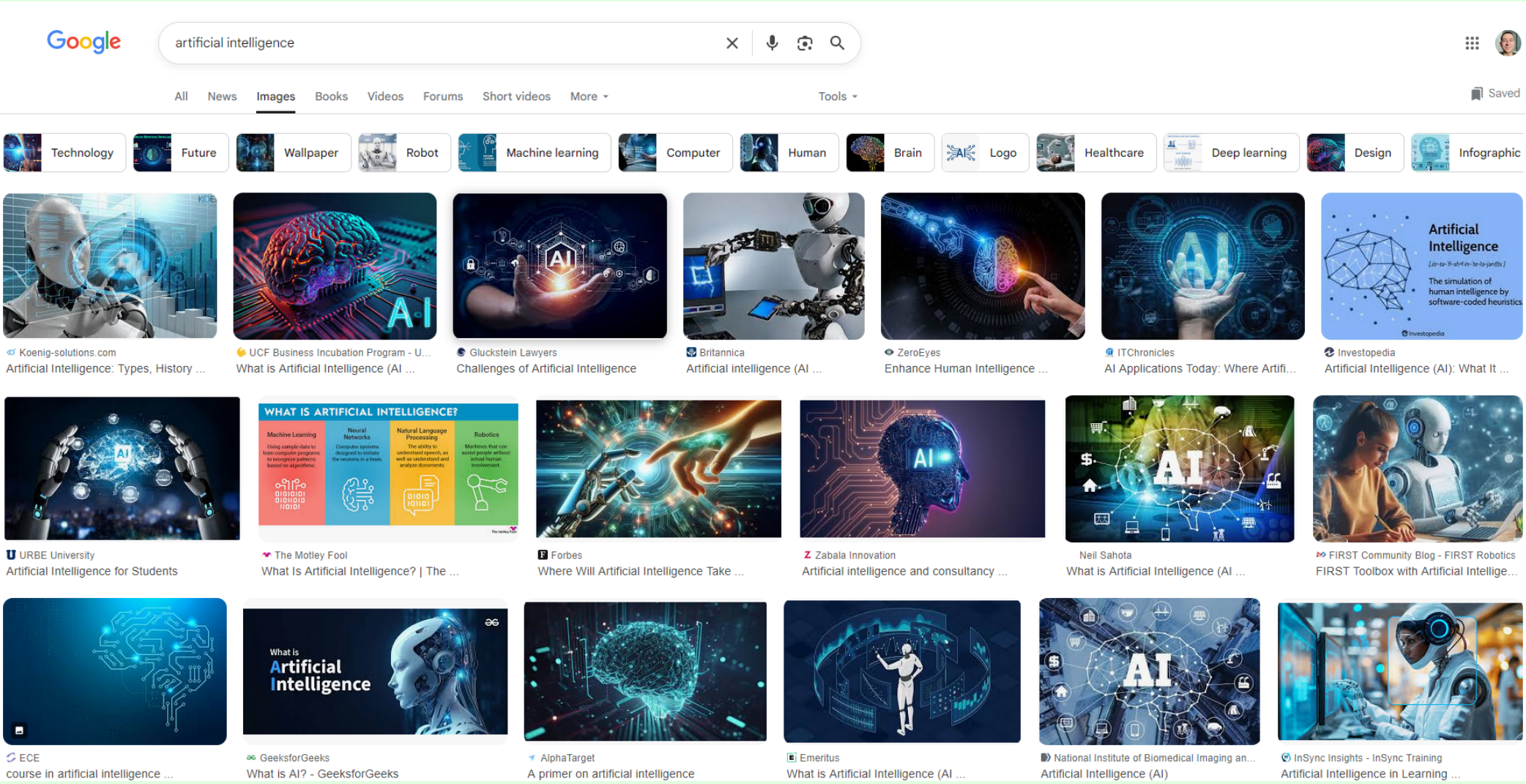
Artificial intelligence

Context, definitions, applications

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What is “artificial intelligence”?



Do you recognize them?









Thinking (?) machines

- Humans tend to recreate **their look** in their works
  - „In the likeness of“: visual arts and sculpture
  - Functional replacements (prostheses)
- Many devices are also designed to imitate human **actions**
  - „Handiwork“ of varying precision (sewing, build line robots, …)
  - Self-moving devices or their parts
  - Simulated communication („speaking“ and “listening”)

- How to determine if any machine can **think** for itself?
  - …what does „thinking“ even mean?

Interpretations of AI

Behaves human-like	**Reasons human-like**
Turing test	Cognitive science Neural networks etc.
Behaves rationally	**Reasons rationally**
Rational agents Problem-solving algorithms	Knowledge representation Systems of formal logic

Turing test for intelligence

- A. M. Turing, *Mind* **59** (236), 433 (1950)
  - [Consider] the "**imitation game**".
    - It is played with three people, a man, a woman, and an interrogator.
    - The interrogator stays in a room apart from the other two.
    - The object of the game for the interrogator is to determine which is the woman.
    - It is [man's] object in the game to try and cause (...) the wrong identification.
    - The object of the game for the third player is to help the interrogator.
  - **What will happen when a machine takes the part of [the man]?**
    - Will the interrogator decide wrongly as often when the game is played like this as he does when the game is played between a man and a woman?
  - These questions replace our original, "**Can machines think?**"

Turing test: discussion

- Thought expression through language(s)
  - Does Englishman stop „thinking“ while in Chile on vacation?
- Physical limitations and societal conventions
  - Most computers can calculate far faster than humans
  - Should they purposely idle before answering math question?
- Black-box estimation
  - Test does not care how thought works – should it?
- …so what constitutes „thought“ in **humans**?

Cognitive science

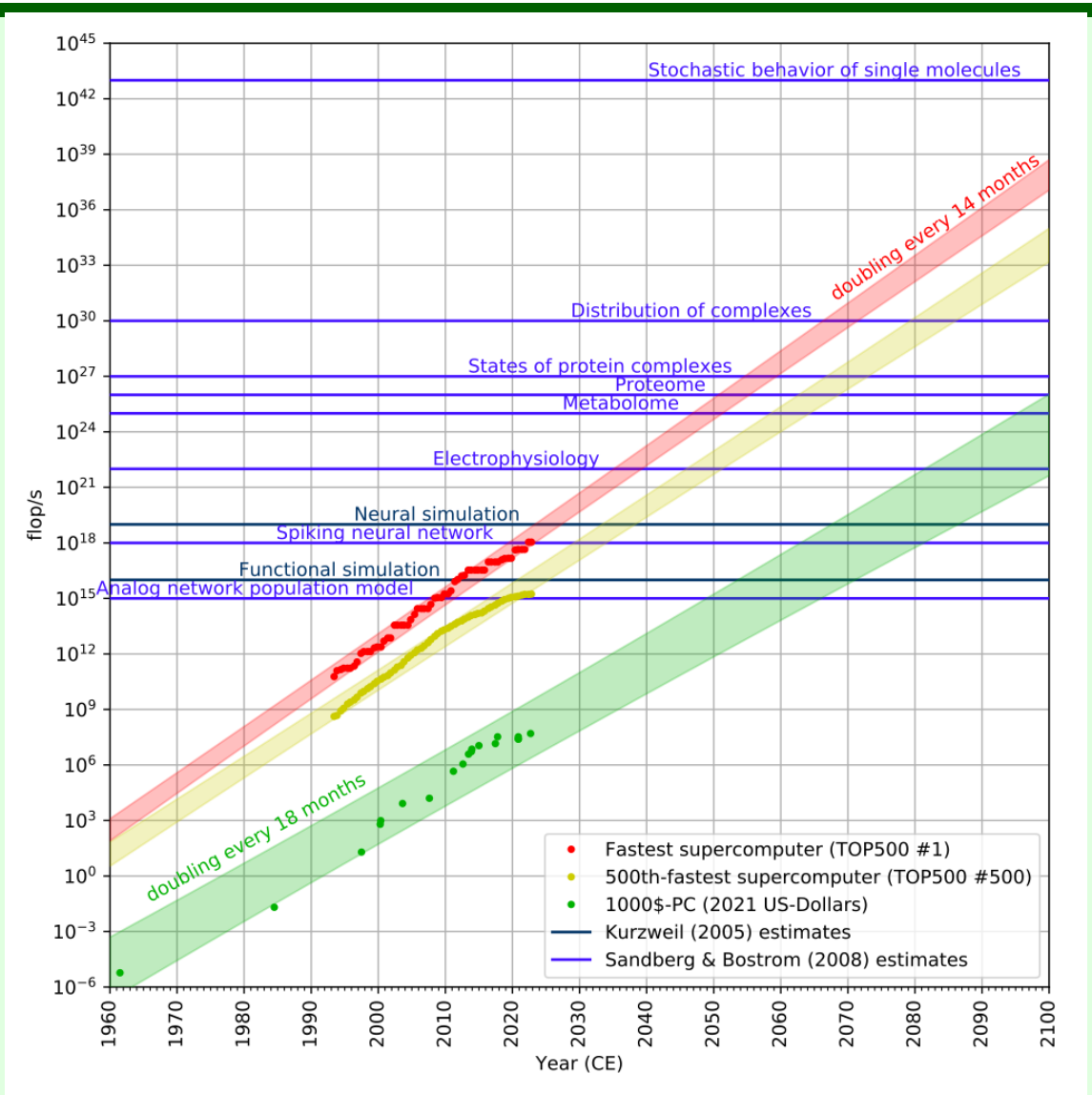
- Physiological and psychological aspects of thought
- Physiology:
  - Synapses and neurons
  - Central and peripheral nervous system
  - Specific brain areas and their functions

- Psychology:
  - Senses, voluntary and involuntary reactions
  - Attention, focus, recall
  - Short- and long-term memory
  - Learning processes and techniques

Cognitive science: discussion

- Human brain is… a *very* complex system
- Do you want to “be” a roundworm? (2015)
  - *https://github.com/Flowx08/Celegans-simulation*
- Fruit fly brain is also available (~~2014~~ **update**: *Nature*, 2024)
  - Shiu *et al*., ***DOI***: 10.1038/s41586-024-07763-9
- Young rat neocortex (2015)
  - H. Markram *et al*., ***DOI***: 10.1016/j.cell.2015.09.029
  - Conflicting reports for the whole mouse brain as of 2025, but we have the scans at least (Johnson *et al*., ***DOI***: 10.1073/pnas.2218617120)
- Technical challenges of human brain modeling
  - J. Jordan *et al*., ***DOI***: 10.3389/fninf.2018.00002, /fninf.2014.00078
  - J. Igarashi, ***DOI***: 10.1016/j.neures.2024.11.005

Computational load estimates



Cognitive science: discussion (2)

- Body-mind dichotomy
- Simulation of neural matter might be too mechanistic
  - Bound to repeat evolutionary mistakes along with breakthroughs
- Psychology, on the other hand, is very hard to quantify
  - Main tool – human dialog (with all its pitfalls)

- …is it possible to think/ **reason** without brain?

Knowledge representation

- Skills and knowledge can be stored in two ways:
  - Implicitly (training, display) → *procedural knowledge*
  - Explicitly (spoken and written word) → *declarative knowledge*
- Declarative knowledge is expressed by *statements*
  - Statement can describe *term* (object) or *formula* (relation) of symbols
  - System of axioms and rules used to join symbols is *formal logic*
- "Thought" in this case is ***logical inference***
  - Behavior based on logical conclusions is usually called *rational*

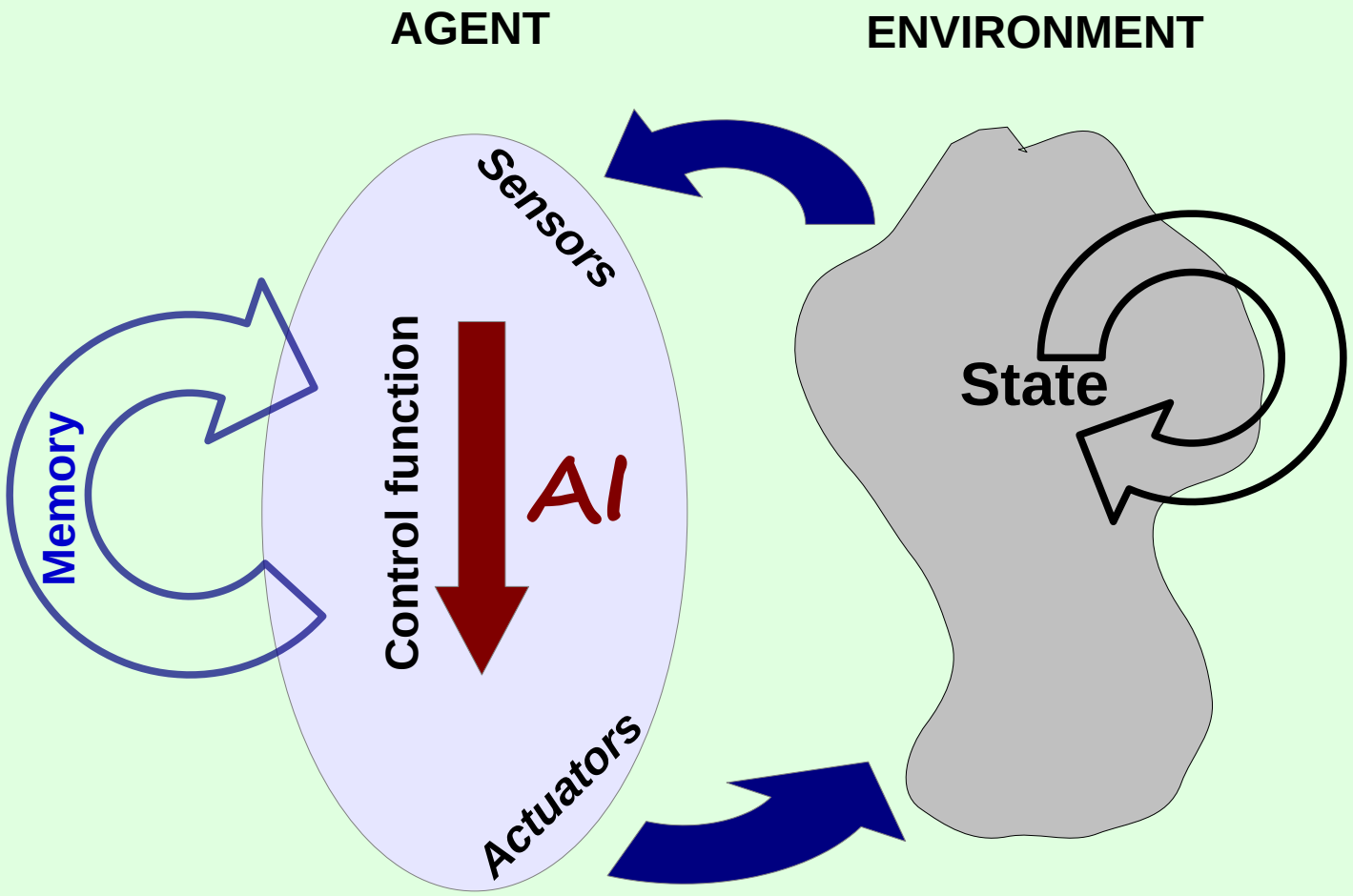
Knowledge representation: discussion

- Not all knowledge is easy to encode
  - What is „common sense“? What is intuition?
  - How to apply inference to incomplete input data?
- Chinese room thought experiment
  - Let’s say I am in a room in a library (oriental studies wing)
  - I am provided with enough Chinese textbooks…
  - …**and** a set of rules on how to use them to adequately „respond“ to various Chinese symbols written on a paper sheet put under the door
  - Now, from the point of outside observer, who actually **knows** Chinese?
    - Argument can be made for a) the rule-set, b) myself or c) the room as a whole
- …and finally: should it even matter?

Rational (intelligent) agent

- A system that is able to effectively **reach goals**
  - Goals can be externally provided or self-formulated
  - In common speech, “effective” and “rational” also tend to go together
- In this case, method of solving becomes less important
  - *A connectivist* will design a neural network
  - *A symbolist* might cook up or use a theorem solver
  - *An engineer* will probably shape, analyze and optimize available data
  - (...and a „vibe coder“ might use the results of all of that)
- A single **agent** can combine different methods

Agent in principle



Note on the usage of AI / ML / DL

- Artificial intelligence – broad field of concepts
  - Subject of this course
- Machine learning – set of computing techniques
  - Part of this course
- ANNs / deep learning – specific structure(s) of solver(s)
  - Small part / applied solution variant for this course
- LLMs – specific subtype of ANNs

Course material

- *Artificial Intelligence: a Modern Approach*
  - S. Russell, P. Norvig (2010 / **2020**) – ***AIMA***
- *Fundamentals of the New Artificial Intelligence*
  - T. Munakata (2008) – ***NewAI***
- **Main focus** – on the concepts and algorithms of the „agents“
  - Some programming knowledge is essential
  - Main language – *Python*

Course evaluation

- Homework assignments (40 %)
  - Some graded, some brought back to discuss in class
- Presentation (20 %)
  - You’re “teaching” others on a given topic
- Project assignment (40 %)
  - Typically in pairs

- Final exam – only if not content with the cumulative grade
  - Exam replaces part of the grade

Course coverage

- State-space search
- Probabilistic reasoning
- Machine learning
- Logic and planning
- Utility and strategy
- Adversarial puzzles and games
- Large language models (?)
- Applications: computer vision, robotics, natural language, ...
- Artificial neural networks
- Genetic algorithms/ programming
- Fuzzy logic

Back to topic: types of agents

- Reflexive
  - Executes a fixed set of rules that bind sensor input and actuator output
- Model-based
  - Stores internal model of the world, updates it through observations
  - Actions are chosen from *current state of the model*
- Planning
  - Analyzes chains of observations and actions
  - Searches for *best available state of the model*
- Effective (utility-based)
  - *Optimizes* its actions in uncertain (partially defined) environment
- Fully independent
  - *Formulates* and solves tasks for itself

Agent environment

- A set of objects to formulate the goals in
  - Usually *system* denotes agent, and *environment* – its playing field
  - Confusingly, an environment is also sometimes called „system“
    - Something for an agent to „solve“, „fix“ or „optimize“

- Environment can be:
  - *Physical* (a room, a highway, a desert)
  - *Idealized* (conveyor belt, board game)
  - *Conceptual* (internet, corpus of texts, set of axioms)

Environment properties

- Fully / partially observable
  - How many observations are needed to *define current state*?
- Deterministic / stochastic
  - Are the results of agent's actions *fully predictable*?
- Discrete / continuous
  - Do states and their elements form *countable set*?
- Static / dynamic
  - Can a state change *without agent's actions*?
- Neutral / adverse / friendly
  - How (and if) it reacts to agent's *internal goals*?

Environment properties: an example

- **Please classify**: checkers, poker, darts, self-driving car

	Observable	Deterministic	Discrete	Static	Neutral
Checkers	V	V	V	V	X
Poker	X	V	V	V	X
Darts	V/X	X	V/X	V/X	V
Car	X	X	X	X	V

"Hell is other people"

- Does a piece of environment seem to have own goals?
  - *Adversaries* act to diminish effectiveness of agent's actions
  - *Partners* act to help an agent reach its goals
  - *Neutral* agents can often be abstracted out
- Additional things to consider:
  - *Communication* between different agents
  - *Metamodels* (modeling of other agents' models)

AI as uncertainty management

- What to do if not sure what to do?
- Limitations of *agents*
  - Amount/ variety/ quality of sensors and actuators
- Limitations of *models*
  - Approximations of varying order
- Stochastic environment
  - Randomness, noise, chaotic behavior
- Limited resources
  - Solving time, computation hardware, accessible materials